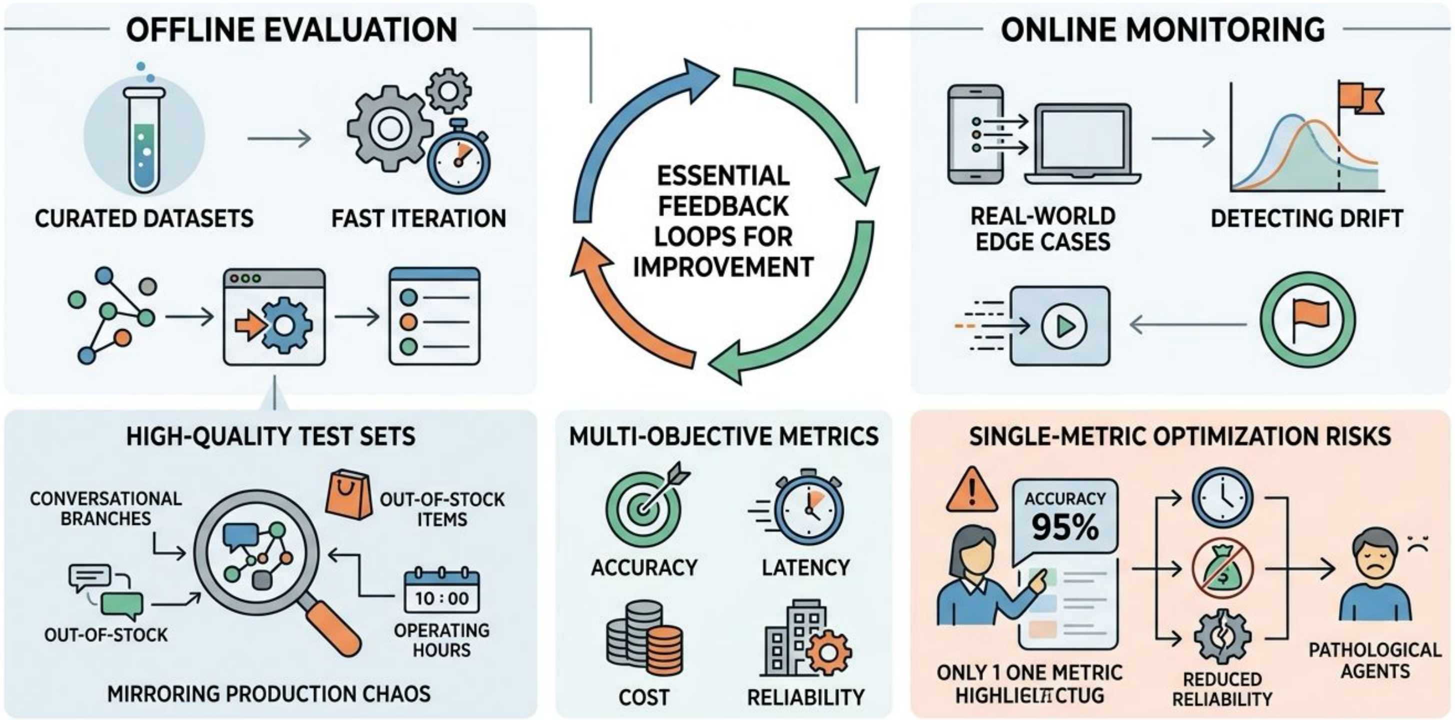
U.S. General Services Administration

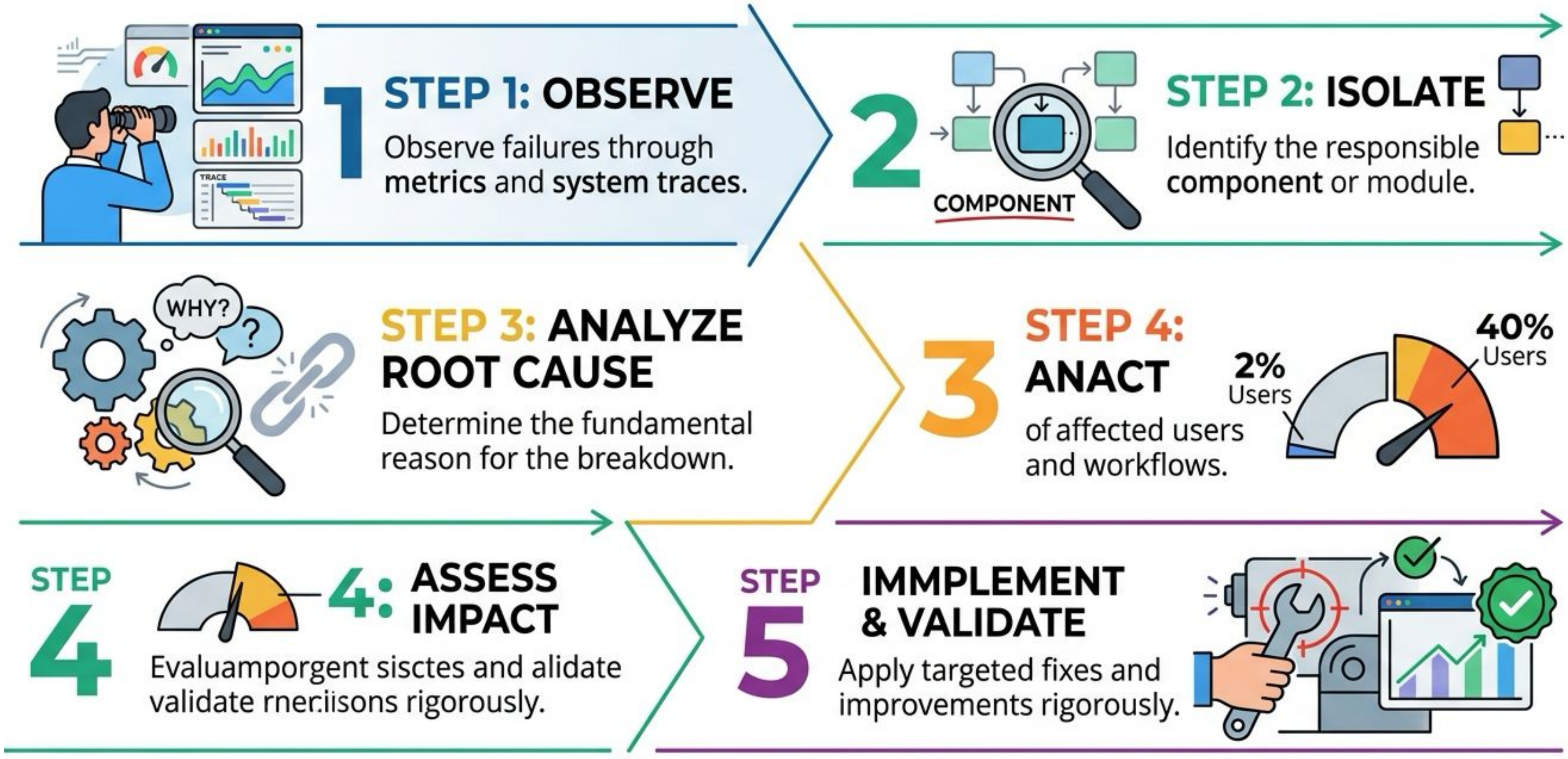
Chapter 3.5: Prompt Optimization, Few-Shot Learning, and Fine-Tuning

US General Service Administration
AI Community of Practice - March 2026

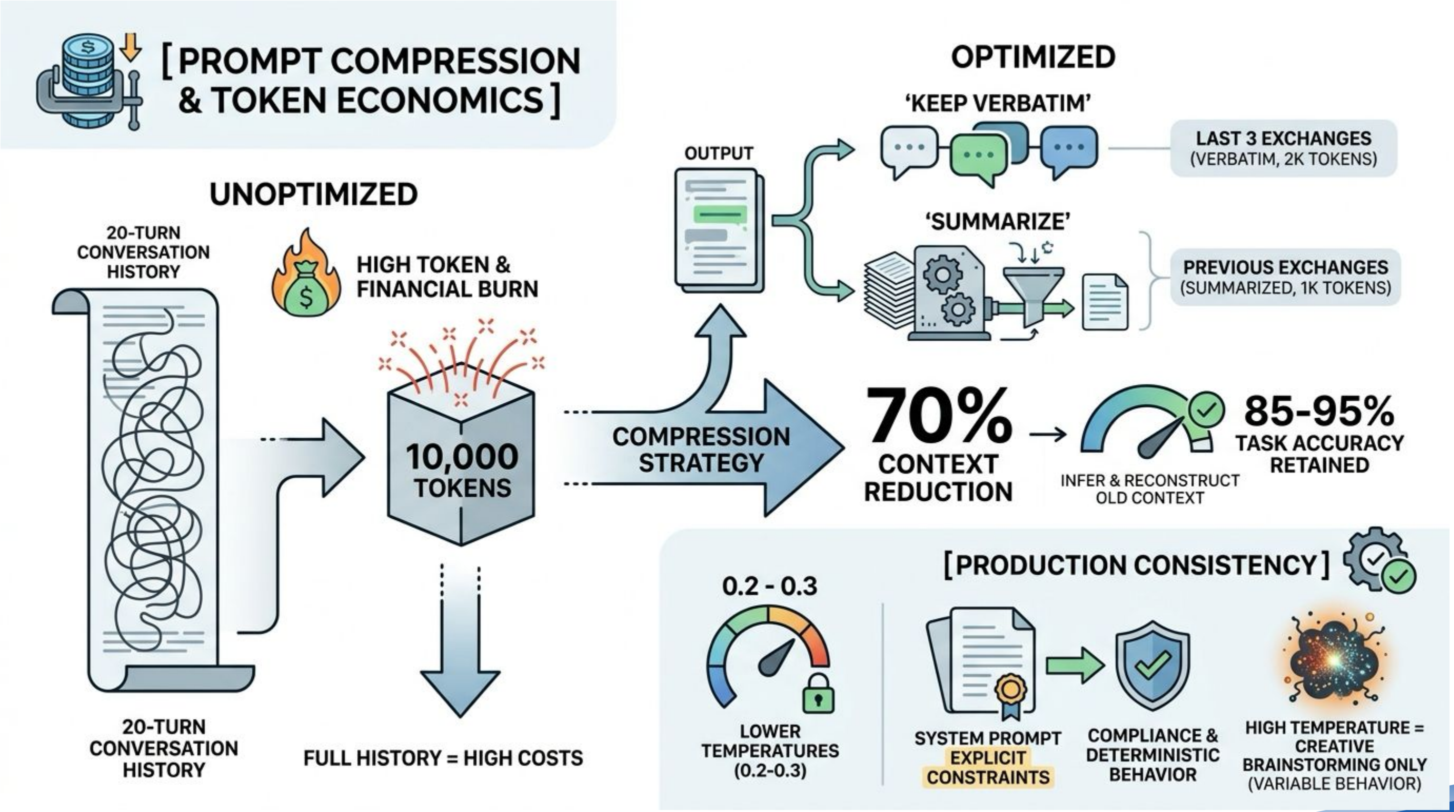
Evaluation Pipelines as Optimization Foundations



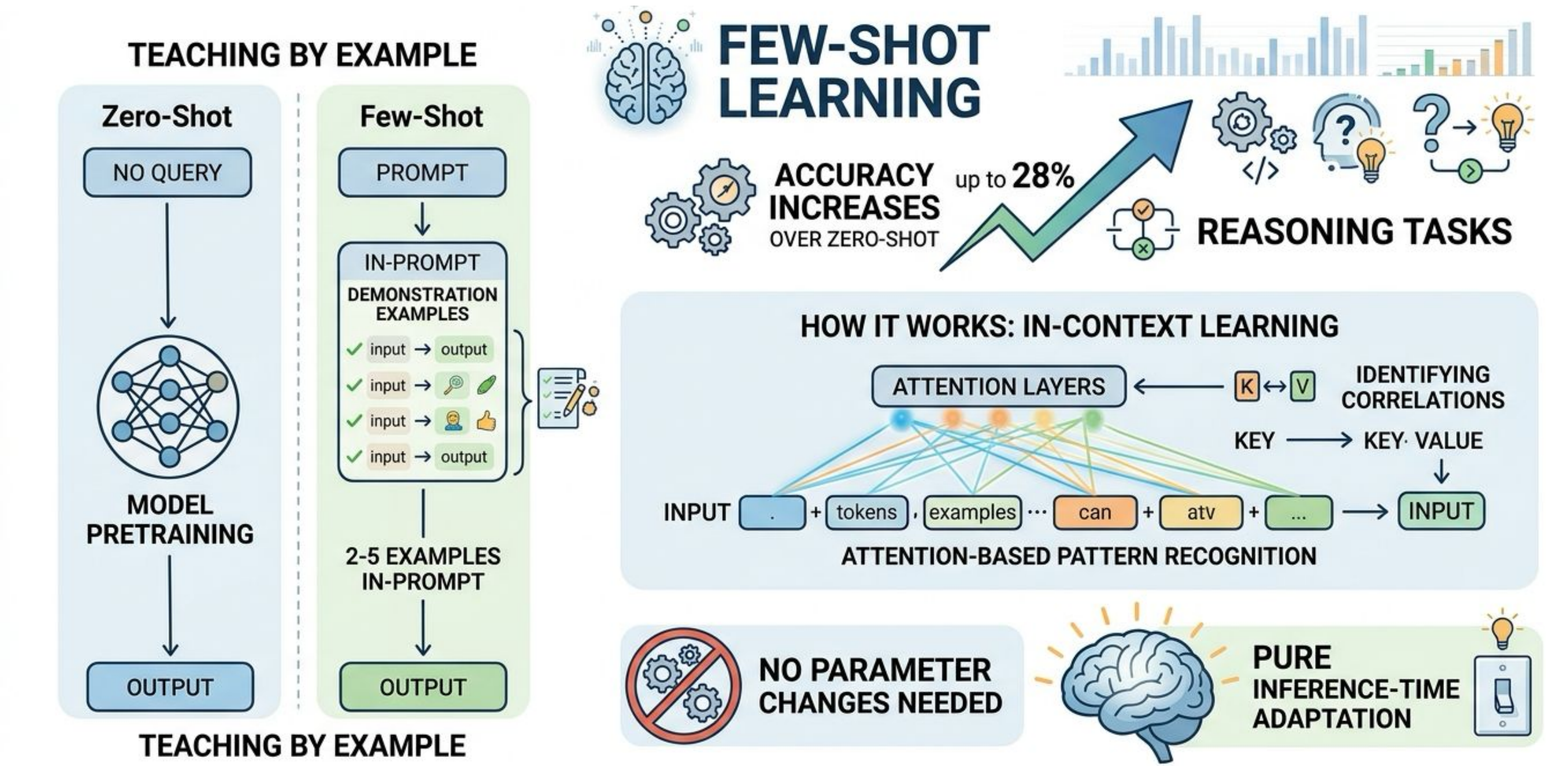
Systematic Optimization -- The Five-Step Diagnostic



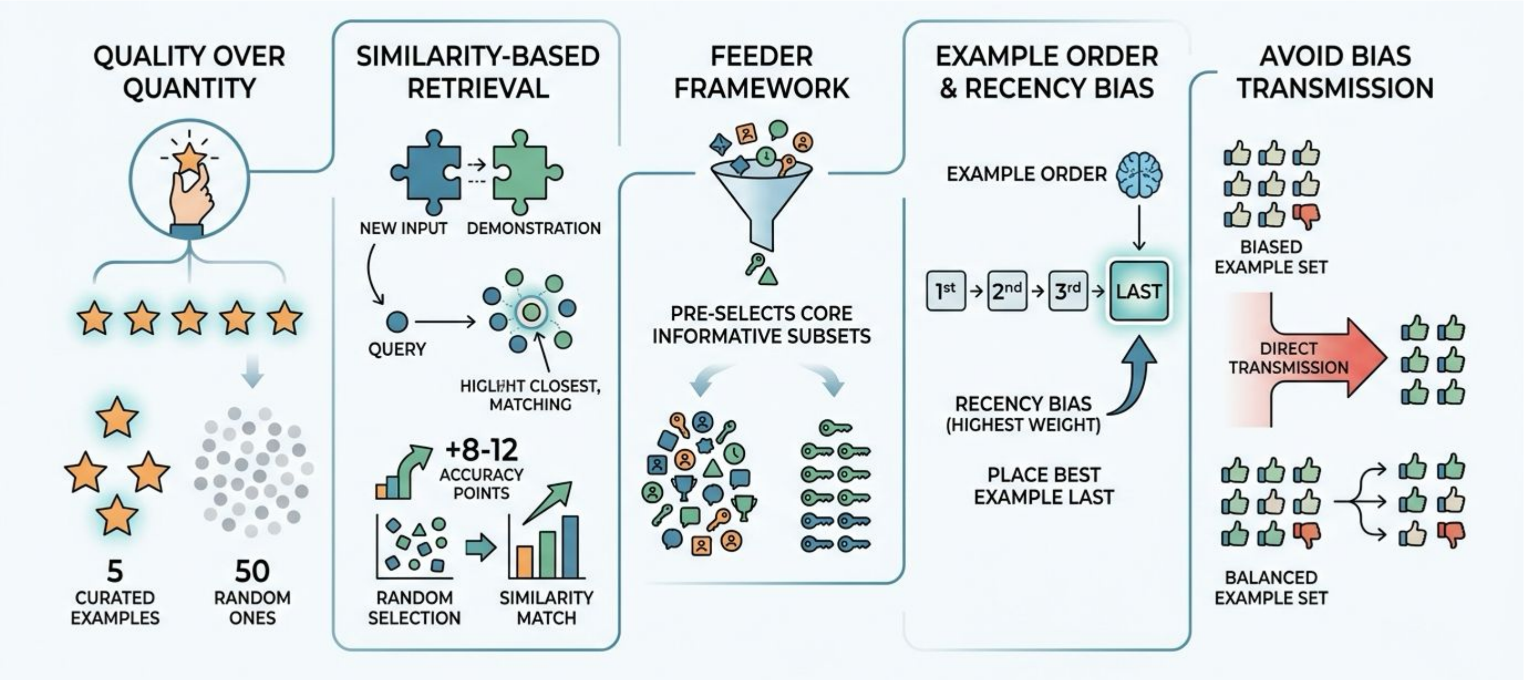
Prompt Compression and Token Economics



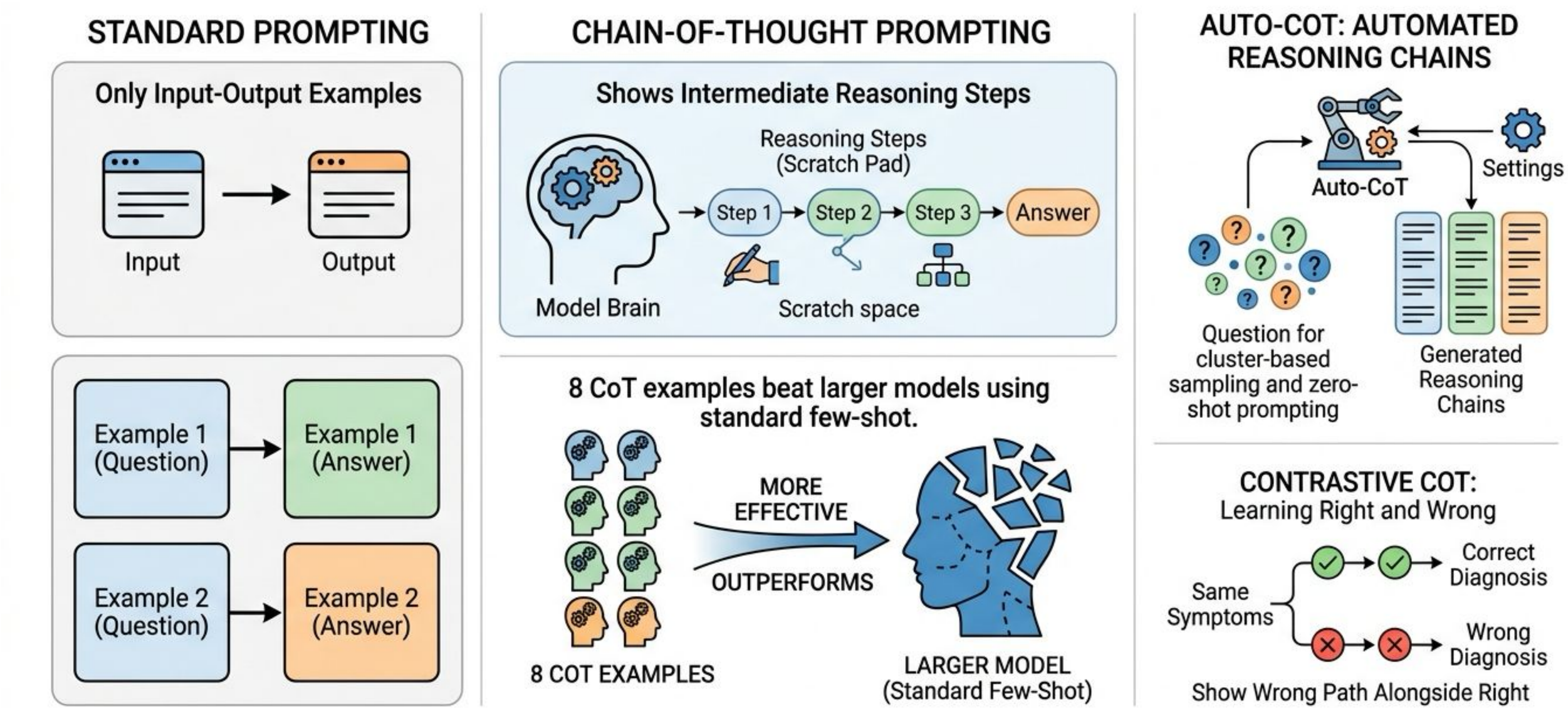
Few-Shot Learning -- Teaching by Example



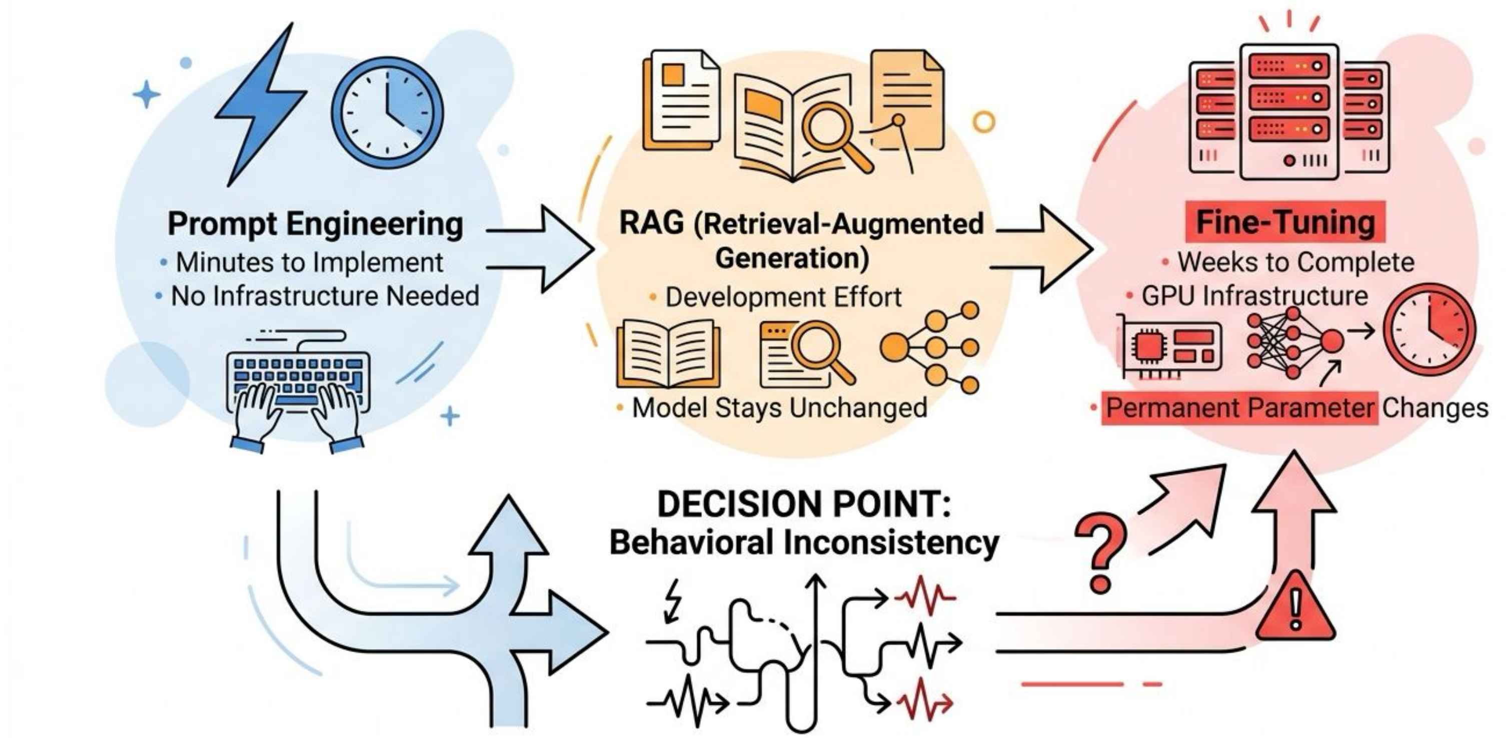
Demonstration Quality and Selection Strategies



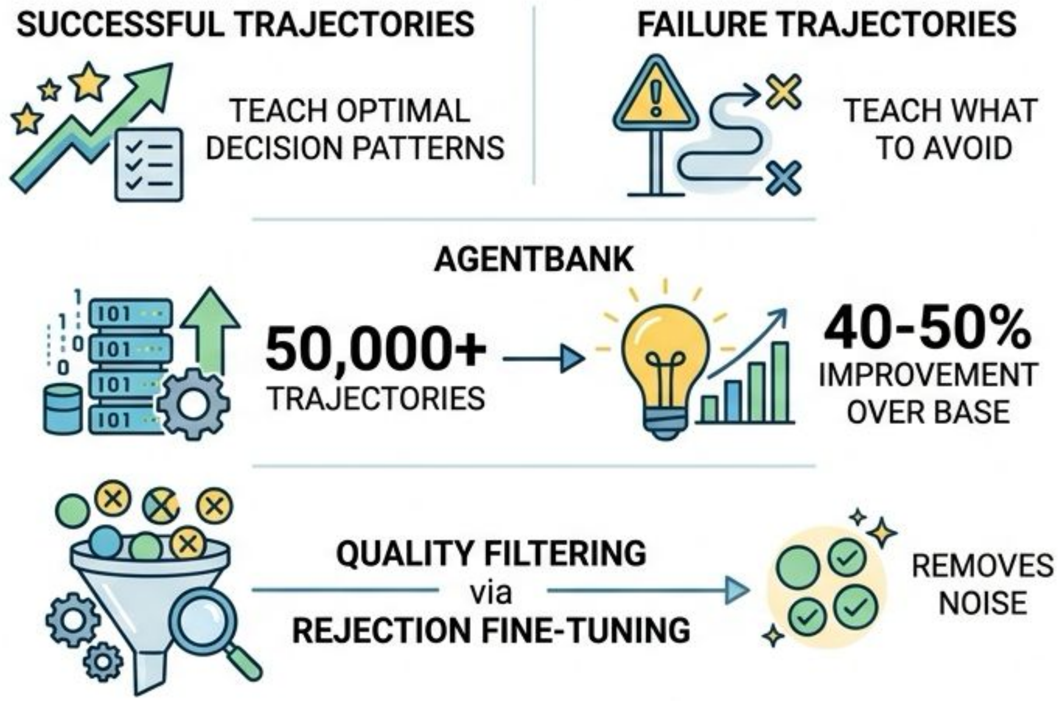
Chain-of-Thought Prompting for Reasoning



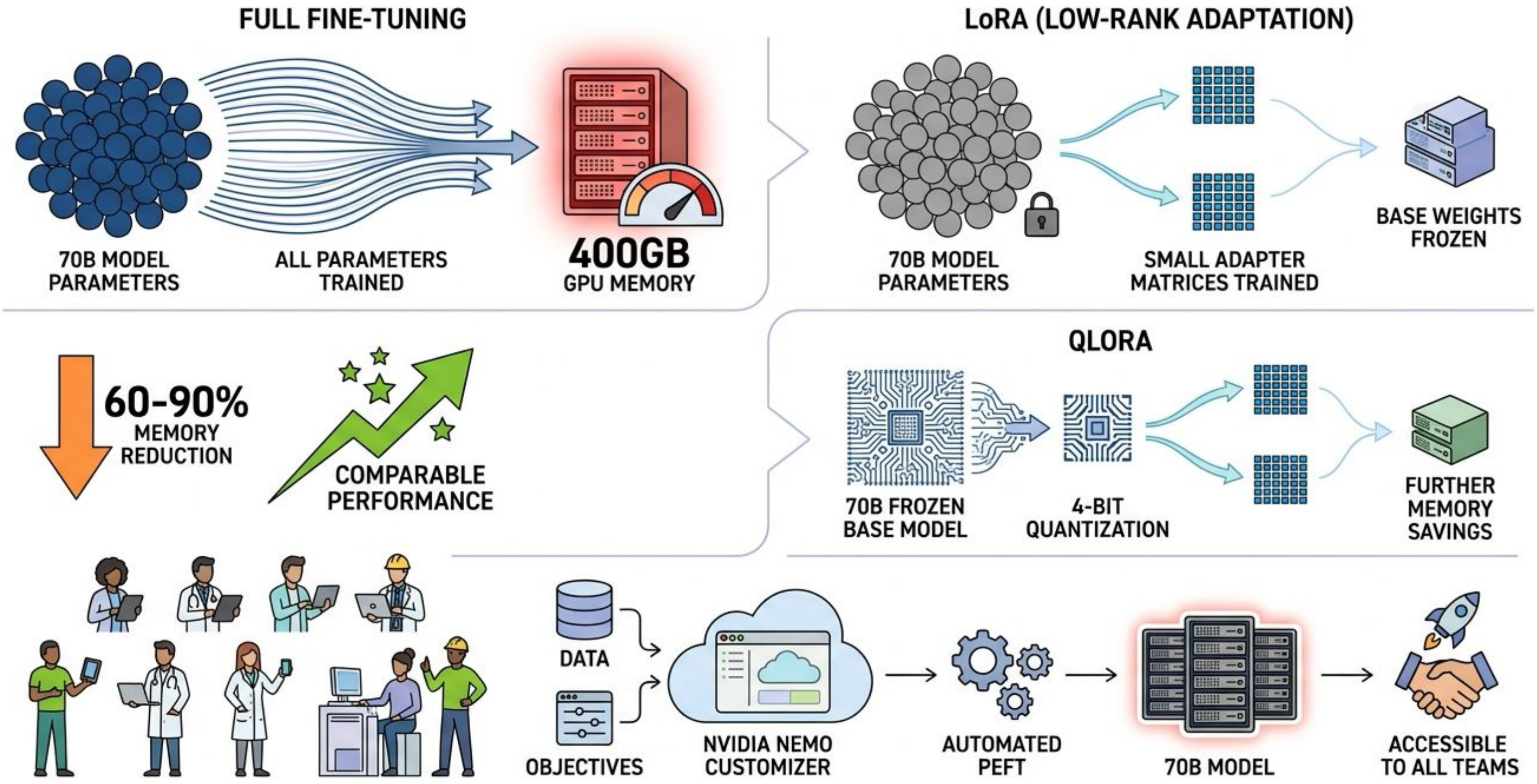
When to Escalate from Prompting to Fine-Tuning



Learning from Agent Trajectories



Parameter-Efficient Fine-Tuning with LoRA



RLHF -- Aligning Agents to Human Preferences

- Some behaviors resist explicit demonstration
- RLHF learns from comparative human judgments
- Stage 1: SFT baseline; Stage 2: reward model; Stage 3: RL
- DPO simplifies to two stages, reducing compute
- Beware annotation biases: length, confidence, fluency

Key Takeaways

- Prompt optimization is engineering, not art -- measure everything
- Few-shot examples: quality and order beat quantity
- Fine-tune only after prompts and RAG reach their ceiling
- PEFT democratizes fine-tuning; RLHF captures preferences
- Next: trace analysis reveals why agents fail step-by-step